Edwin Lotsu

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Objective

Master's student in Computer Engineering at Clemson University with hands-on experience in SIEM-based security monitoring, network intrusion detection, and malware reverse engineering. Built a Wazuh SIEM lab with attack simulation, custom detection rules, and endpoint telemetry analysis. Proficient in threat detection, log analysis, and security operations. Seeking a Summer 2026 SOC Analyst or cybersecurity internship.

Education

Clemson University, Clemson, SC **Present**
MSc, Computer Engineering
- **Expected Graduation:** December 2026
- **Relevant Coursework:** Distributed Denial of Services Attacks (DDOS), Computer Network Security, Malware Reverse Engineering

Ghana Communication Technology University, Ghana **2023**
BSc, Computer Engineering

Experience

Enscript Solutions Ltd **Nov 2023 - Dec 2024**
Software Engineer (National Service) *Accra, Ghana*
- Designed and implemented a USSD-based fintech payment platform with secure mobile-to-backend connections, including role-based access control, API token management, and multi-factor authentication across 3-tier architecture, protecting sensitive financial data from unauthorized access.
- Implemented input validation and sanitization controls across REST API endpoints to mitigate injection attack vectors and enforce data integrity on payment transactions.
- Collaborated with project managers and QA team to deliver client-facing fintech products on schedule and resolve reported bugs.

Immigration Services, Clemson University **Jan 2025 - Present**
Graduate Assistant *Clemson, SC*
- Managed systems access provisioning and compliance workflows in Sunapsis, ensuring appropriate access controls and secure handling of student records in accordance with institutional policy.
- Developed technical documentation and internal training materials covering systems processes, contributing to international students and staff onboarding and compliance awareness.
- Supported date reporting and IT operations using Excel and SharePoint, managing institutional datasets with accuracy and attention to data integrity.

Technical and Coursework Projects

Network Security Labs & Malware Reverse Engineering
- Built a comparative IDS pipeline using Snort and Suricata; evaluated detection rule effectiveness against evasion techniques, analyzed false positive rates using Wireshark traffic captures, and assessed trade-offs between detection sensitivity and alert noise.
- Performed static and dynamic analysis on malware using IDA Pro, Process Monitor, Process Explorer, and PE Viewer to identify attack vectors, persistence mechanisms, and indicators of compromise (IOCs).
- Simulated network attack scenarios including buffer overflow exploitation, and packet sniffing; analyzed encrypted traffic behavior under VPN tunneling to assess confidentiality and detection visibility.
- **Tools:** IDA, Process Monitor, Virtual Private Networks, Wireshark, Linux, Snort, Suricata

Security Monitoring and Threat Detection Home Lab
- Built a multi-VM security operations lab using Wazuh SIEM to monitor Windows Server (Active Directory), domain-joined Windows 10 endpoint, and Ubuntu infrastructure; deployed Sysmon for enhanced endpoint telemetry and simulated attack scenarios including network reconnaissance (Nmap port scanning) while analyzing detection artifacts through SIEM dashboards and structured queries.
- Developed custom detection rules to identify suspicious network activity patterns and privilege escalation attempts; mapped alerts to MITRE ATT&CK techniques including Network Service Scanning (T1046) and validated detection coverage through offensive simulations
- **Tools**: Wazuh, Sysmon, Kali Linux, Nmap, Windows Server (Active Directory), Ubuntu, VMware

Skills

- **Programming**: Python, JavaScript
- **Security & Analysis Tools**: Wireshark, Snort, Suricata, IDA Pro, Process Monitor, VPN, Malware Static & Dynamic Analysis, Windows Server (Active Directory, SIEM Lab Deployment)
- **SIEM & Detection**: Wazuh, Custom Detection Rules, MITRE ATT&CK Mapping, Log Analysis, Threat Hunting
- **Soft Skills**: Technical Writing, Security Documentation, Communication, Team Collaboration